

Integrated Circuit ECE326

Lec04

Synthesis Digital Circuit

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☐ Synthesize of CMOS digital circuit

☐ Exercise_1

☐ Exercise_2

☐ Exercise_3

☐ Exercise_4



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Synthesize of CMOS digital circuit (1/5)



Ex.1. Synthesize $F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{C}$ using
(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) CMOS transistors

Step 1: $F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{C}$ is simplified by K-map

		B			
		00	01	11	10
A	0	1	0	1	1
	1	1	1	0	1

BC is indicated by a red diagonal line. A blue bracket under the 01 and 11 columns is labeled 'C'. A blue bracket under the 11 and 10 columns is labeled 'B'.

Step 2: $\overline{F} = A \cdot B \cdot C + \overline{A} \cdot \overline{B} \cdot C = C \cdot (A \cdot B + \overline{A} \cdot \overline{B})$

Synthesize of CMOS digital circuit (2/5)



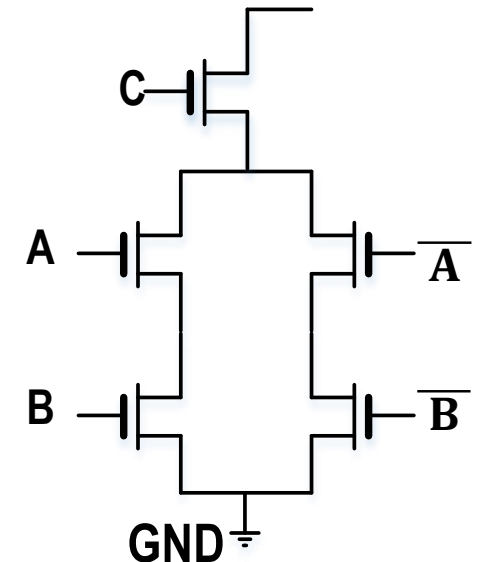
Ex.1. Synthesize $F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{C}$ using
(a) CMOS transistors (b) Universal gates (c) Compare the used transistors
in (a) and (b)

Ans.
(a) CMOS transistors

Step 2: $\overline{F} = C \cdot (A \cdot B + \overline{A} \cdot \overline{B})$

Step 3: Firstly, draw the pull-down network

- ☐ Two transistors $\{A, B\}$ in series connection
- ☐ Two transistors $\{\overline{A}, \overline{B}\}$ in series connection.
- ☐ $\{A, B\}$ and $\{\overline{A}, \overline{B}\}$ are connected in parallel connection.
- ☐ Transistor $\{C\}$ is in series connection with $\{A, B\}$ and $\{\overline{A}, \overline{B}\}$



Synthesize of CMOS digital circuit (3/5)



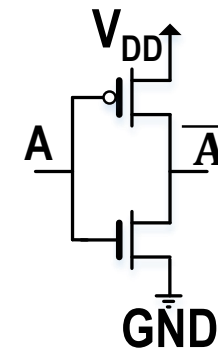
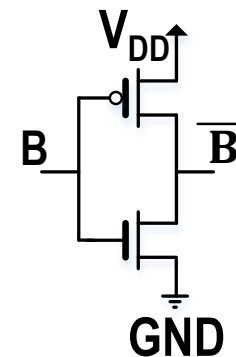
Ex.1. Synthesize $F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{C}$ using
(a) CMOS transistors (b) Universal gates (c) Compare the used transistors
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Ans.
(a) CMOS transistors

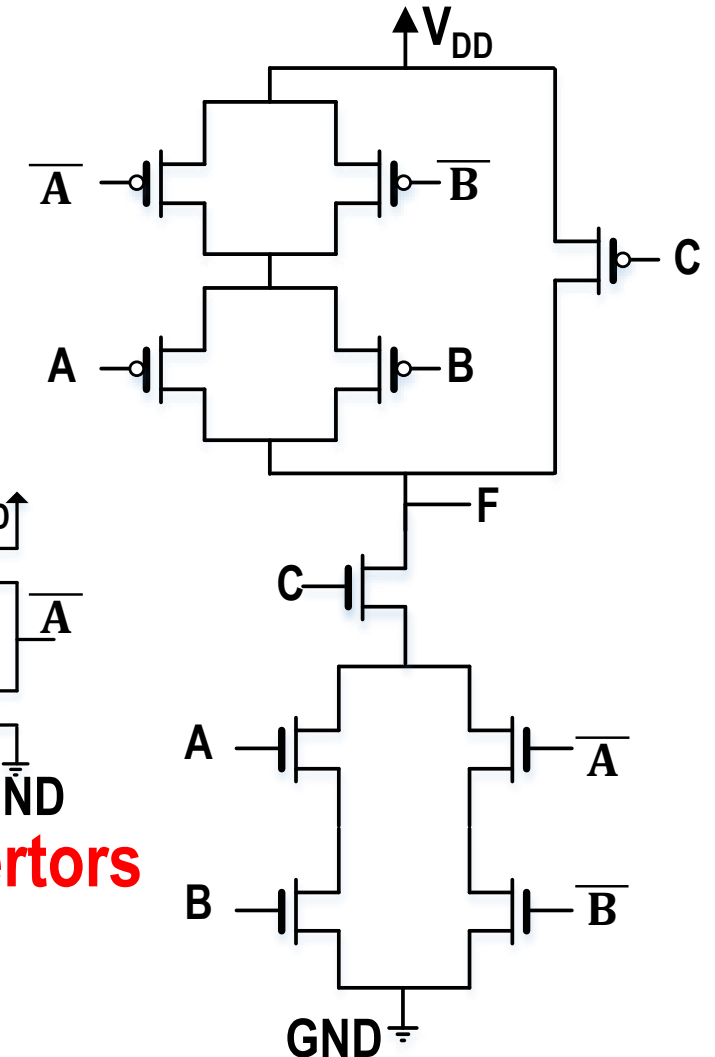
Step 2: $\overline{F} = C \cdot (A \cdot B + \overline{A} \cdot \overline{B})$

Step 4: Secondly, draw the pull-up network

- ❑ Two transistors {A,B} in parallel connection
- ❑ Two transistors { $\overline{A} \cdot \overline{B}$ } in parallel connection.
- ❑ {A,B} and { $\overline{A} \cdot \overline{B}$ } are connected in series connection.
- ❑ Transistor {C} is in parallel connection with {A,B} and { $\overline{A} \cdot \overline{B}$ }



Don't Forget the invertors



of transistors = $2 \cdot 5 + 2 \cdot 2 = 14$ 😊

Synthesize of CMOS digital circuit (9/11)



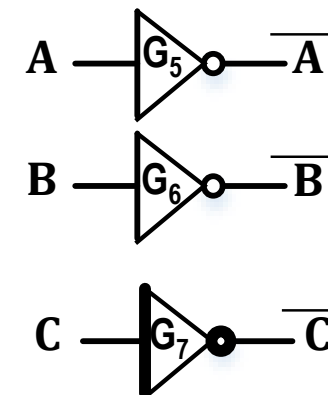
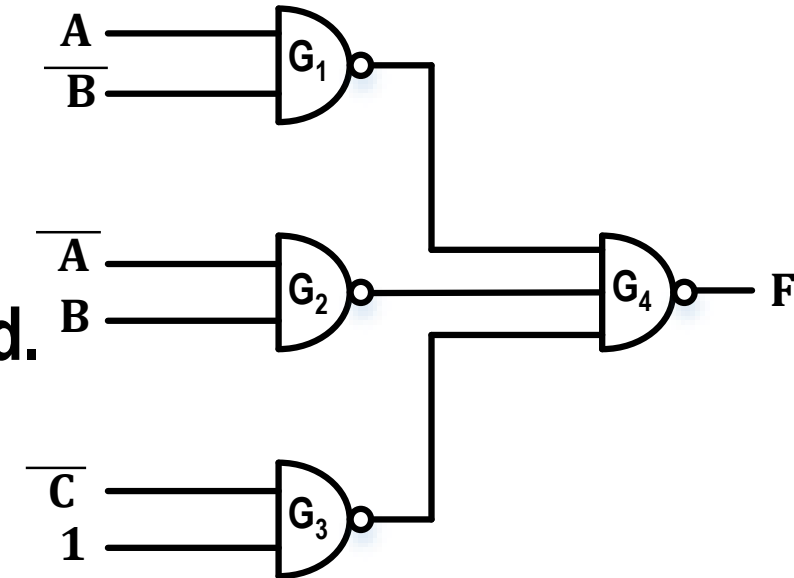
Ex. Synthesize $F = A \cdot \bar{B} + \bar{A} \cdot B + \bar{C}$ using

(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) Universal gates

Due to the function is in form of **SOP**, NAND gates are used.

$$\begin{aligned}\overline{\overline{F}} &= \overline{A \cdot \bar{B} + \bar{A} \cdot B + \bar{C} \cdot 1} \\ \overline{\overline{F}} &= \overline{\underbrace{A \cdot \bar{B}}_{G_1} + \underbrace{\bar{A} \cdot B}_{G_2} + \underbrace{\bar{C} \cdot 1}_{G_3}} \\ &\quad \underbrace{\hspace{10em}}_{G_4}\end{aligned}$$



of transistors = $3 \cdot 2 \cdot 2 + 1 \cdot 3 \cdot 2 + 3 \cdot 2 = 24$ 😊

Synthesize of CMOS digital circuit (10/11)



Ex. Synthesize $F = A \cdot \bar{B} + \bar{A} \cdot B + \bar{C}$ using

(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.

(C) Compare the used transistors in (a) and (b)

	CMOS transistors	Universal gates
# of transistors	14 😊	24 ☹️

Synthesizing by **CMOS transistors** requires low number of transistors,
Thus, the consumption power is low and the consumed silicon area is small



☐ Synthesize of CMOS digital circuit

☐ Exercise_1

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Synthesize of CMOS digital circuit (1/5)



Ex.2. Synthesize the function $f(A,B,C, D) = \overline{(A + B + C)} \cdot D$

(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) CMOS transistors

Step 1: $F = \overline{(A + B + C)} \cdot D$ is simplified directly

Step 2: $\overline{F} = (A + B + C) \cdot D$

You don not
need to use
K-Map

Note

You don not need to use **K-Map**

If you can get
 $\overline{F}$ **directly**

If the function has **> 4**
variables, it is **hard** to
use K-Map

Synthesize of CMOS digital circuit (2/5)



Ex.2. Synthesize the function $f(A,B,C, D) = \overline{(A + B + C)} \cdot D$

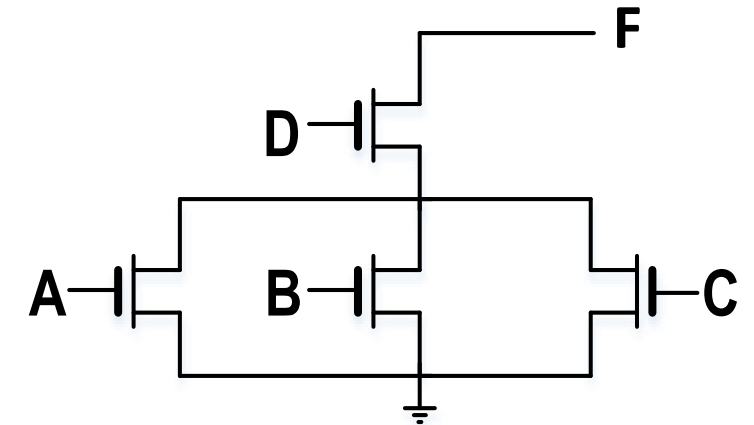
(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) CMOS transistors

Step 2: $\overline{F} = (A + B + C) \cdot D$

Step 3: **Firstly**, draw the **pull-down** network

- Three transistors {A,B,C} are connected in parallel connection
- Then three transistors {A,B,C} are connected **with** {D} in series connection.



Synthesize of CMOS digital circuit (3/5)



Ex.2. Synthesize the function $f(A,B,C, D) = \overline{(A + B + C)} \cdot D$

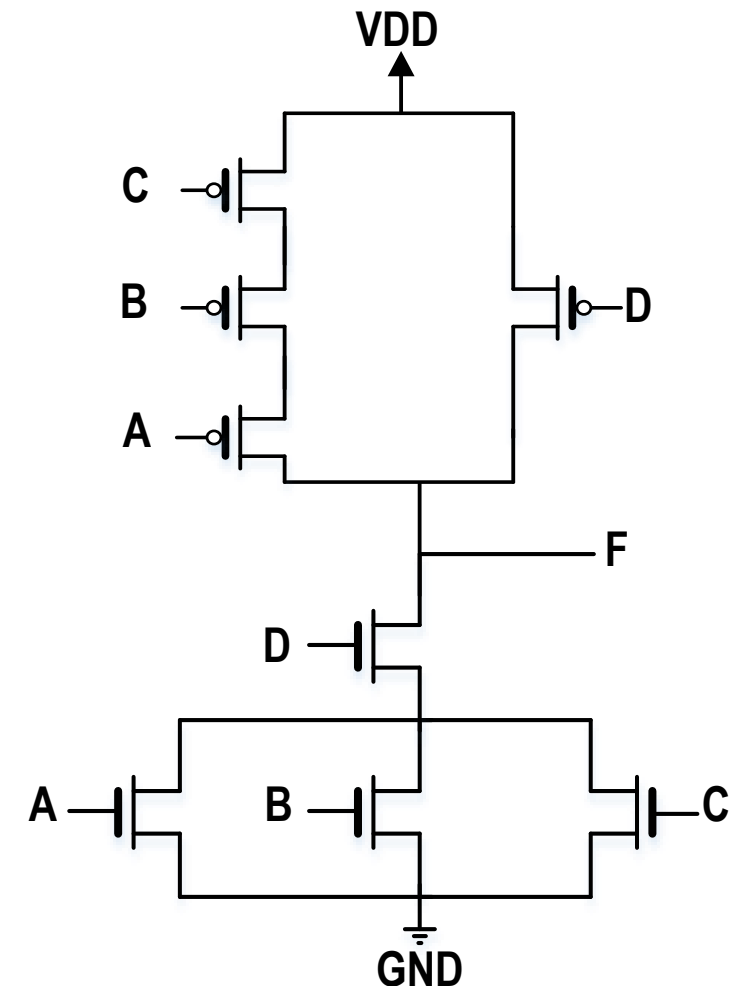
(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) CMOS transistors

Step 2: $\overline{F} = (A + B + C) \cdot D$

Step 4: **Secondly**, draw the **pull-up** network

- Three transistors {A,B,C} are connected in series connection
- Then, three transistors {A,B,C} are connected **with** {D} in parallel connection



of transistors= 2*4= 8



Synthesize of CMOS digital circuit (4/5)

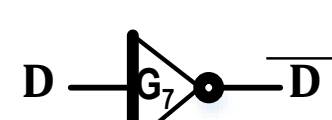
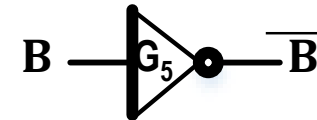
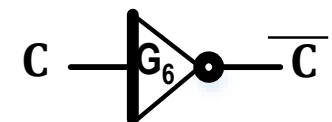
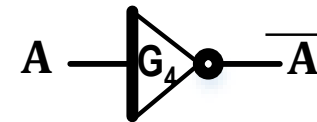
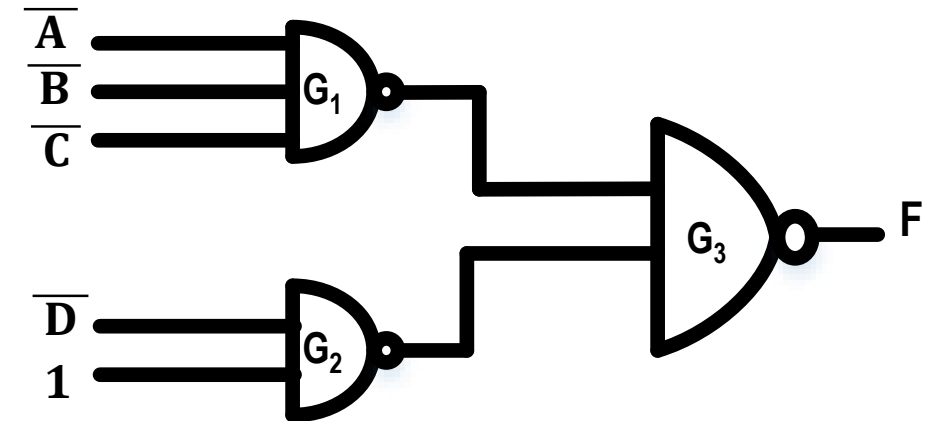


Ex.2. Synthesize the function $f(A,B,C, D) = \overline{(A + B + C)} \cdot D$

(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) Universal gates

$$F = \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{D}$$
$$\overline{\overline{F}} = \overline{\overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{D}}$$
$$\overline{\overline{F}} = \underbrace{\overline{\overline{A} \cdot \overline{B} \cdot \overline{C}}}_{G_1} \cdot \underbrace{\overline{\overline{D} \cdot 1}}_{G_2} \quad G_3$$



of transistors = $1 \cdot 3 \cdot 2 + 1 \cdot 2 \cdot 2 + 1 \cdot 2 \cdot 2 + 4 \cdot 2 = 22$ 😊

Synthesize of CMOS digital circuit (5/5)



Ex.2. Synthesize the function $f(A,B,C, D) = \overline{(A + B + C)} \cdot D$

(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.

(C) Compare the used transistors in (a) and (b)

	CMOS transistors	Universal gates
# of transistors	8 😊	22 ☹️

Synthesizing by CMOS transistors requires low number of transistors,
Thus, the consumption power is low and the consumed silicon area is small



☐ Synthesize of CMOS digital circuit

☐ Exercise_1

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☐ Exercise_4

Synthesize of CMOS digital circuit (1/5)



Ex.3. Design a three input odd-parity checker using minimum number of transistor .Noted that the output is high when an odd number of 1's.

Ans.

Step 1: Build the truth table to get the function



The output function is obtained by using Sum Of Product (**SOP**)

$$F = A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot B \cdot \overline{C} + \overline{A} \cdot \overline{B} \cdot C + A \cdot B \cdot C$$

Input			Output
C	B	A	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Synthesize of CMOS digital circuit (2/5)



Ex.3. Design a three input odd-parity checker using minimum number of transistor .Noted that the output is high when an odd number of 1's.

Ans.

Step 1: Build the **truth table** to get the function

$F = A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot B \cdot \overline{C} + \overline{A} \cdot \overline{B} \cdot C + A \cdot B \cdot C$ is simplified by **K-map**

		B			
		00	01	11	10
A	0	0	1	0	1
	1	1	0	1	0

Step 2: $\overline{F} = \overline{A} \cdot \overline{B} \cdot \overline{C} + A \cdot \overline{B} \cdot C + \overline{A} \cdot B \cdot C + A \cdot B \cdot \overline{C}$

$$\overline{F} = \overline{A} \cdot (\overline{B} \cdot \overline{C} + B \cdot C) + A \cdot (B \cdot \overline{C} + \overline{B} \cdot C)$$

Synthesize of CMOS digital circuit (3/5)



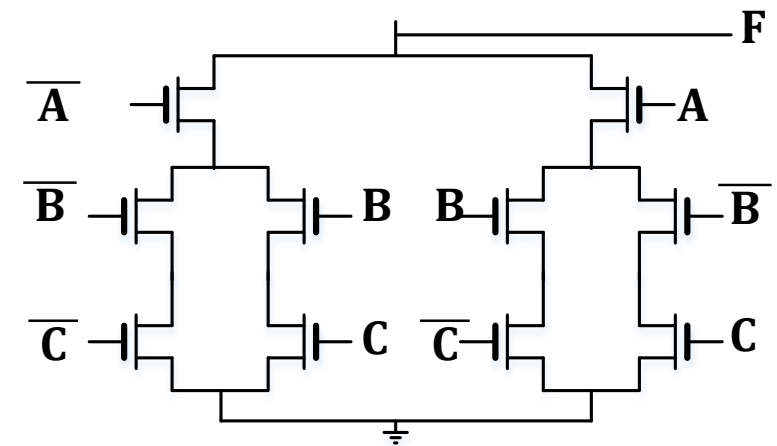
Ex.3. Design a three input odd-parity checker using minimum number of transistor .Noted that the output is high when an odd number of 1's.

Ans.

Step 2: $\overline{F} = \overline{A} \cdot (\overline{B} \cdot \overline{C} + B \cdot C) + A \cdot (B \cdot \overline{C} + \overline{B} \cdot C)$

Step 3: Firstly, draw the pull-down network

- ☐ Two transistors $\{\overline{B} \cdot \overline{C}\}$ are connected in series connection
 - ☐ Two transistors $\{B \cdot C\}$ are connected in series connection
 - ☐ $\{\overline{B} \cdot \overline{C}\}$ & $\{B \cdot C\}$ are connected in parallel
 - ☐ Then , all $\{\overline{B} \cdot \overline{C}\}$ & $\{B \cdot C\}$ are connected with $\{\overline{A}\}$ in series.
-
- ☐ Two transistors $\{B \cdot \overline{C}\}$ are connected in series connection
 - ☐ Two transistors $\{\overline{B} \cdot C\}$ are connected in series connection
 - ☐ $\{B \cdot \overline{C}\}$ & $\{\overline{B} \cdot C\}$ are connected in parallel
 - ☐ Then , all; $\{B \cdot \overline{C}\}$ & $\{\overline{B} \cdot C\}$ are connected with $\{A\}$ in series



Synthesize of CMOS digital circuit (4/5)



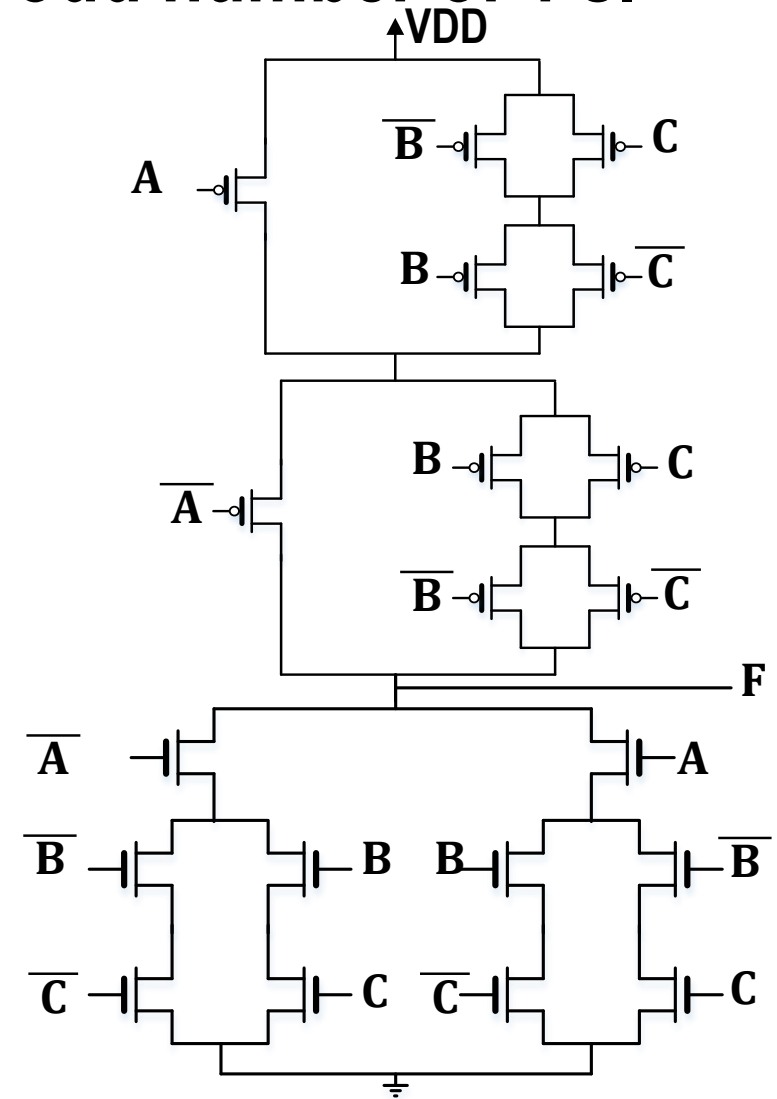
Ex.3. Design a three input odd-parity checker using minimum number of transistor .Noted that the output is high when an odd number of 1's.

Ans.

Step 2: $\overline{F} = \overline{A} \cdot (\overline{B} \cdot \overline{C} + B \cdot C) + A \cdot (B \cdot \overline{C} + \overline{B} \cdot C)$

Step 4: Secondly, draw the pull-up network

- ☐ Two transistors $\{\overline{B} \cdot \overline{C}\}$ are connected in **parallel** connection
 - ☐ Two transistors $\{B \cdot C\}$ are connected in **parallel** connection
 - ☐ $\{\overline{B} \cdot \overline{C}\}$ & $\{B \cdot C\}$ are connected in **series**
 - ☐ Then, all $\{\overline{B} \cdot \overline{C}\}$ & $\{B \cdot C\}$ are connected with $\{\overline{A}\}$ in **parallel**.
-
- ☐ Two transistors $\{B \cdot \overline{C}\}$ are connected in **Parallel** connection
 - ☐ Two transistors $\{\overline{B} \cdot C\}$ are connected in **Parallel** connection
 - ☐ $\{B \cdot \overline{C}\}$ & $\{\overline{B} \cdot C\}$ are connected in **series**
 - ☐ Then , all $\{B \cdot \overline{C}\}$ & $\{\overline{B} \cdot C\}$ are connected with $\{A\}$ in **parallel**

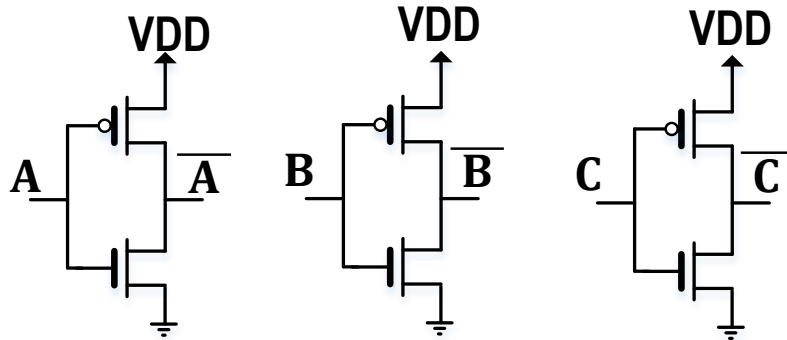


Synthesize of CMOS digital circuit (5/5)



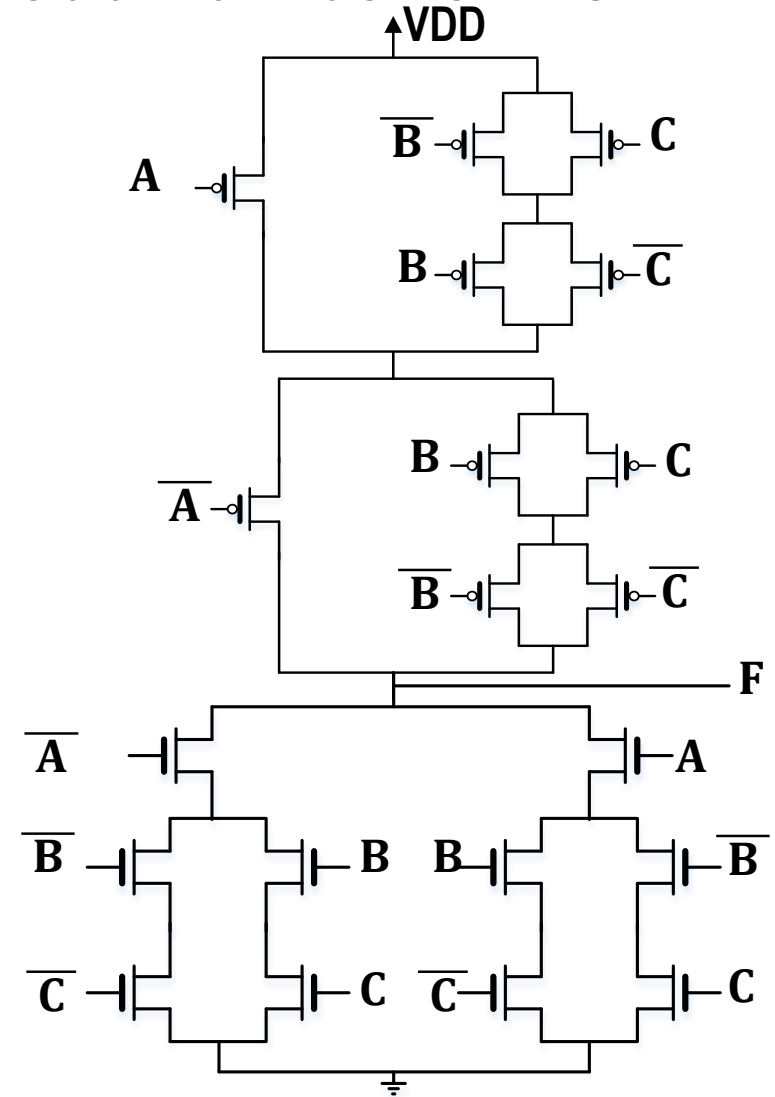
Ex.3. Design a three input odd-parity checker using minimum number of transistor .Noted that the output is high when an odd number of 1's.

Ans.



Don't Forget the inverters

of transistors = $10 \times 2 + 3 \times 2 = 26$ 😊





☐ Synthesize of CMOS digital circuit

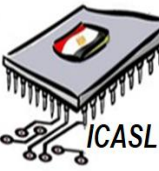
☐ Exercise_1

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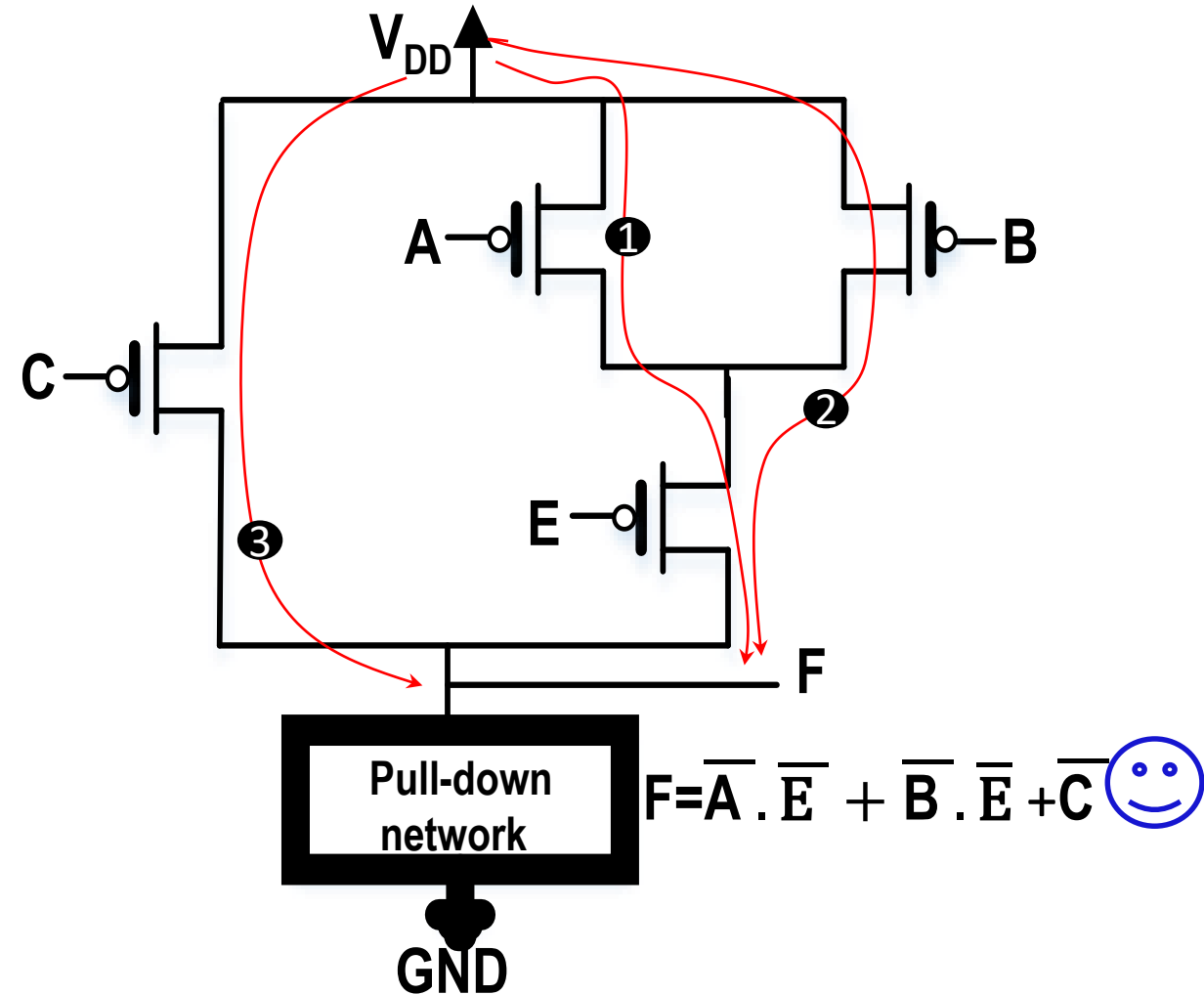
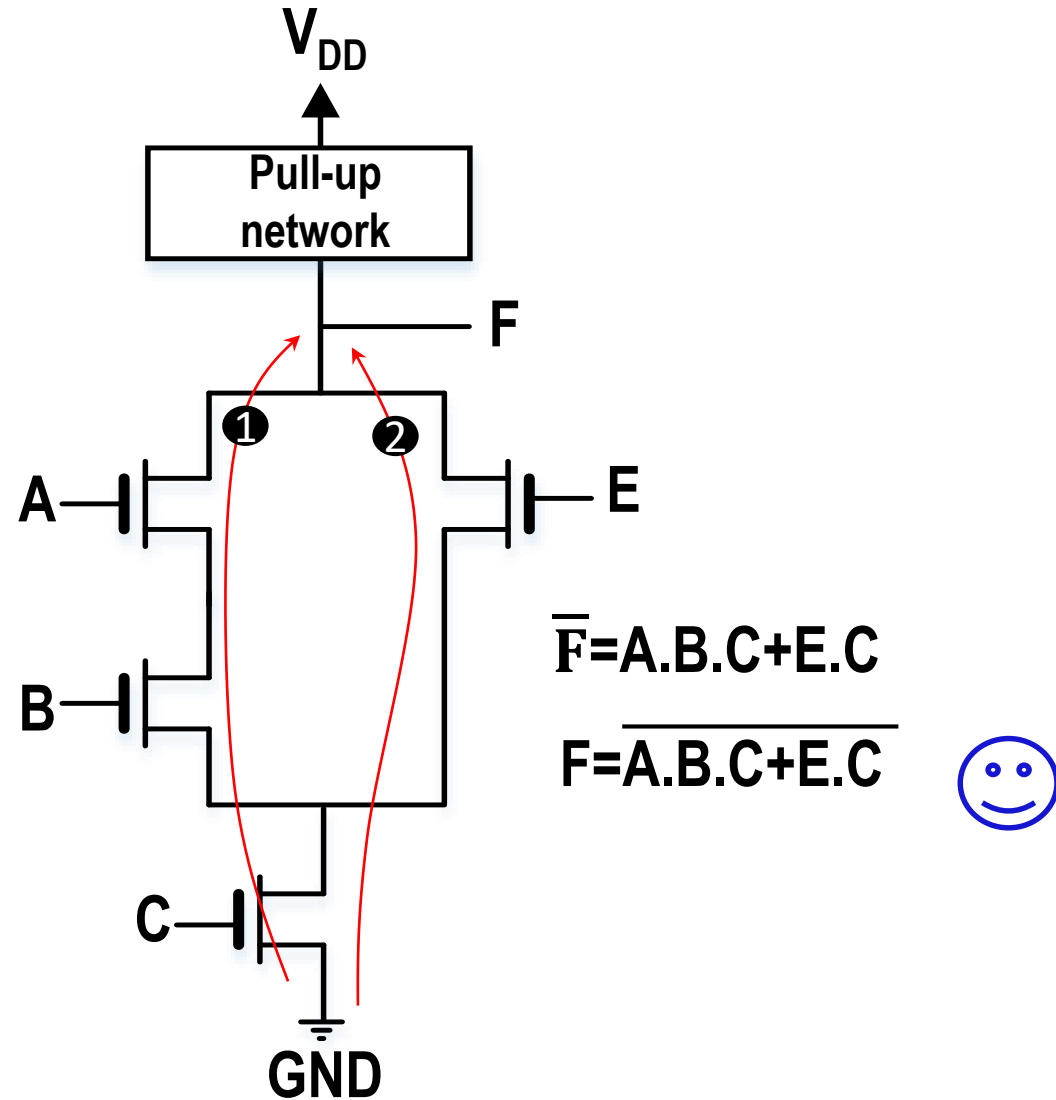
☐ Exercise_3

☐ Exercise_4

Synthesize of CMOS digital circuit



Ex.4. Find the transfer function of the following circuit



Thanks!